



CERTIFICATE OF PARTICIPATION

This is to certify that

John Michael Jabuen

successfully presented the paper titled

"A Multithreaded Multi-Colony Ant Colony Optimization Framework with Ring and Random Communication Topologies for Solving Large-Scale Sudoku"

at the 6TH International Conference on

NOVEL & INTELLIGENT DIGITAL SYSTEMS (NIDS) 2026

held at the University of West Attica, Athens, Greece, September 23–25, 2026.

On behalf of NiDS2026 Organizing Committee

A. Krouska

*Asst. Prof. Akrivi Krouska
Program Co-Chair*